

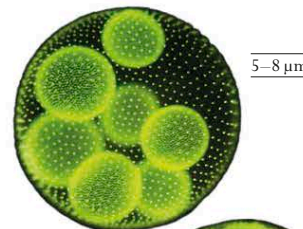
GREEN ALGAE

Green algae are a loose taxonomic aggregation of very diverse species. Some live in freshwater ponds and streams, some form green mats over damp, shady rocks or tree trunks, and others grow as leafy seaweeds, attached to rocks in shallow sea water.

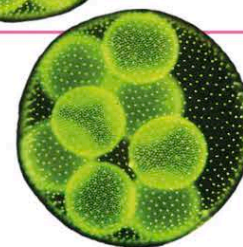
Many green algae are microscopic and free-floating; others are multicellular with variably complex structures, including the blanketweeds

that sometimes clog ponds. Some reproduce asexually by splitting, budding, or producing motile spores, but sexual reproduction is common in larger species. These typically produce sperm with two identical flagella, but also have spore-producing stages in their life cycle. The cells of green algae use the same chlorophyll pigments as land plants for photosynthesis, and they are sometimes included with them in an informal grouping called the Viridiplantae (green plants).

DOMAIN	EUKARYOTA
KINGDOM	PROTOCTISTA
DIVISION	CHLOROPHYTA
FAMILIES	127
SPECIES	4,300



5–8 μm



VOLVOX AUREUS
Just visible to the naked eye, spherical colonies of this freshwater alga consist of thousands of microscopic individuals. Thread-like flagellae spin the colony through the water.

12–60 cm
4¾–23½ in

SEA LETTUCE

Ulva lactuca

A popular food worldwide, this green seaweed has broad, crumpled fronds, anchored to rock by a holdfast. It can also form floating communities.

5–40 cm
2–16 in

GREEN SEA FINGERS

Codium fragile

This tubular green alga occurs worldwide, in seashore rock pools and coastal waters to 2 m (6½ ft) deep.

2–5 m
6½–16 ft

SEA GRAPE

Caulerpa lentillifera

This edible seaweed produces many upright stalks from a long, creeping runner. In the Philippines, the succulent stalks are harvested and eaten raw in salads.

STONEWORKS AND RELATIVES

Although related to green algae, the stoneworts and their relatives have more complex structures and more advanced chemistry in their cells, suggesting that they are the ancestors of the true plants.

The Streptophyta (sometimes called Charophyta) includes desmids, which are classed as “conjugating green algae”. These typically have a single cell, divided into two symmetrical semi-cells.

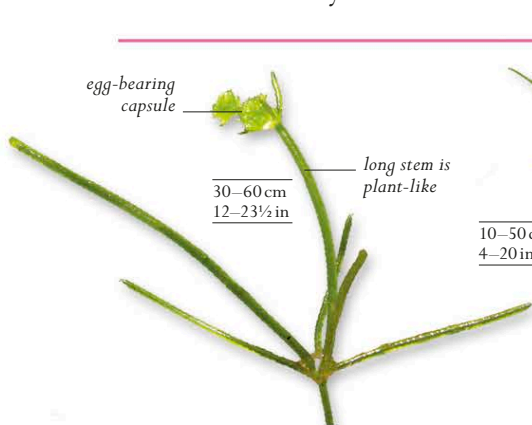
However, the stoneworts and brittleworts are the best known group of streptophytes, and are often regarded as “honorary plants”. They grow in the mud of shallow, freshwater or brackish habitats, rooted by cell filaments, called rhizoids, like the roots of simple plants. Their branched form and reproductive structures are also reminiscent of true plants. Indeed, some classification systems regard the Streptophyta as an “infra-kingdom” that also includes all of the land plants.

DOMAIN	EUKARYOTA
KINGDOM	PROTOCTISTA
DIVISION	STREPTOPHYTA
FAMILIES	16
SPECIES	2,700

egg-bearing capsule

30–60 cm
12–23½ in

long stem is plant-like

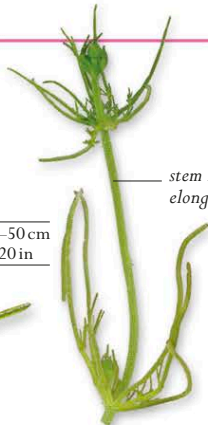


NITELLA TRANSLUCENS

This delicate “plant” with translucent green leaves grows in the clear water of ponds, streams, and fens around W. and S. Europe.

10–50 cm
4–20 in

stem made of elongated cells



COMMON STONEWORT

Chara vulgaris

Also known as skunkweed, this streptophyte of the northern hemisphere emits an unpleasant smell.

32–70 μm



PENIUM SP.

This desmid, found in N. America, is symmetrically divided into cylindrical semi-cells with blunt, oval ends and girdle bands.

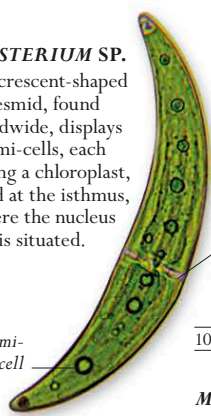
CLOSTERIUM SP.

This crescent-shaped desmid, found worldwide, displays semi-cells, each housing a chloroplast, joined at the isthmus, where the nucleus is situated.

semi-cell

cell divide

100–460 μm



MICRASTERIAS SP.

This desmid, found in temperate areas, has multiple spined arms, tipped with prongs that allow the semi-cells to interlock.

350 μm

prong

